

RET fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes RET gene fusions from the msk_impact_50k_2026 study, covering 194 fusion events. 146/194 fusions (75.3%) were in-frame. 179/194 fusions (92.3%) retained the PF07714 domain. Domain retention was tested with Fisher's exact test ($p=0.000419666$, statistically significant at $\alpha=0.05$). Compared to the literature benchmark (PMCID: PMC6430196 (KIF5B-RET): 100.0% in-frame, 100.0% domain-retained), this run observed 75.3% in-frame and 92.3% domain retention.

1 Results summary

1.1 Fusion partner frequency

Fusion-partner frequency was tabulated across 194 analyzed fusion events, identifying 33 distinct fusion partner genes. The most frequent partner was KIF5B, observed in 87 events.

1.2 Domain retention

PF07714 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 141/146 (96.6%) of in-frame fusions retained the domain, $p=0.000419666$ (statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p -value of 0.000999001.

1.3 Domain disruption

Domain-disruption analysis was skipped: RET has no disruption_required_domains configured; domain-disruption analysis was skipped.

1.4 Cutpoint detection

Cutpoint detection scanned 194 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 627 aa (permutation-corrected $p=0.001998$, statistically significant at $\alpha=0.05$). This is -97 aa from the nearest configured domain boundary at 724 aa.

1.5 Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=179$) and "not_retained" ($n=15$). For Frame_status, retained: 141/157 (89.8%, 95% CI 84.1%-93.6%); not_retained:

5/14 (35.7%, 95% CI 16.3%-61.2%). Welch's t-test comparing Tumor_variant_count between groups gave means of 67.4413 vs 49.2667, $p=0.22222$ (not statistically significant at $\alpha=0.05$).

1.6 Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p -value (e.g. disruption_required_domains not configured for this gene).

1.7 Exon retention

Exon retention reported: No target exon was supplied in params or GeneConfig.

1.8 Joint partner

Joint partner reported: RET has no gene_pair configured; joint-partner analysis was skipped.

2 Tables

2.1 composite_score composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
KIF5B	87	0.4484536082474227	0.30004340774793187		0.8093693978698724	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.5046697811340378	1
CCDC6	51	0.26288659793814434	0.30004340774793187		0.8167835941716136	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.43647231407463494	2
PDCD10	1	0.005154639175257732	0.30004340774793187		1.0	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3741759056313749	3
TFG	1	0.005154639175257732	0.30004340774793187		1.0	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3741759056313749	4
ERC1	1	0.005154639175257732	0.30004340774793187		0.9724770642201834	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.36901535517265927	5
NCOA4	16	0.08247422680412371	0.30004340774793187		0.8031823394495413	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.36626743963898867	6
RBPM5	1	0.005154639175257732	0.30004340774793187		0.9311926605504587	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.36127452948458594	7
RUFY1	2	0.010309278350515464	0.30004340774793187		0.9174311926605505	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.36062724394594975	8
LIN52	1	0.005154639175257732	0.30004340774793187		0.908256880733945	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.35697407076898957	9
EML4	1	0.005154639175257732	0.30004340774793187		0.8944954128440367	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3543937955396318	10
DOCK1	1	0.005154639175257732	0.30004340774793187		0.8371559633027523	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.34364264875064093	11
SPECC1L	4	0.020618556701030927	0.30004340774793187		0.8044724770642202	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3433134641530811	12
TIMM23B	3	0.015463917525773196	0.30004340774793187		0.8035168195718654	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.341201288682543	13
KCNMA1	1	0.005154639175257732	0.30004340774793187		0.823394495412844	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.34106237352128316	14
ERCC6	3	0.015463917525773196	0.30004340774793187		0.8027522935779816	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3410579400586898	15
RUFY3	2	0.010309278350515464	0.30004340774793187		0.805045871559633	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3395549962395278	16
CCDC186	1	0.005154639175257732	0.30004340774793187		0.805045871559633	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.33762200654880614	17
CUBN	1	0.005154639175257732	0.30004340774793187		0.805045871559633	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.33762200654880614	18

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty	Components_applicable	Composite_score	Rank
GEMIN5	1	0.005154639175257732	0.30004340774793187		0.805045871559633	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.33762200654880614	19
KIF13A	1	0.005154639175257732	0.30004340774793187		0.805045871559633	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.33762200654880614	20
RELCH	1	0.005154639175257732	0.30004340774793187		0.805045871559633	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.33762200654880614	21
SLC4A4	1	0.005154639175257732	0.30004340774793187		0.805045871559633	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.33762200654880614	22
CLIP1	1	0.005154639175257732	0.30004340774793187		0.8027522935779816	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.33719196067724644	23
GRIPAP1	1	0.005154639175257732	0.30004340774793187		0.8027522935779816	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.33719196067724644	24
RASSF4	1	0.005154639175257732	0.30004340774793187		0.8027522935779816	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.33719196067724644	25
RUFY2	1	0.005154639175257732	0.30004340774793187		0.8027522935779816	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.33719196067724644	26
SHROOM3	1	0.005154639175257732	0.30004340774793187		0.8027522935779816	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.33719196067724644	27
TRIM24	1	0.005154639175257732	0.30004340774793187		0.8027522935779816	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.33719196067724644	28
KIAA1217	1	0.005154639175257732	0.30004340774793187		0.536697247706422	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.287306639576329	29
CTNNA3	1	0.005154639175257732	0.30004340774793187		0.29587155963302747	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.24215182306256758	30
CTNNA1	1	0.005154639175257732	0.30004340774793187		0.1009174311926605	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.20559792397999876	31
UXS1	1	0.005154639175257732	0.30004340774793187		0.06422018348623848	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.19871719003504462	32
FBXL7	2	0.010309278350515464	0.30004340774793187		0.0	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.18860889532209657	33

2.2 cutpoint_detection cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
219	0	179	1	14	0.0	0.07731958762886598	1.1117104708745449
425	1	178	1	14	0.07865168539325842	0.14903050050745154	0.826724840000421
550	3	176	2	13	0.11079545454545454	0.048844025264081076	1.3111885527954676
556	4	175	2	13	0.14857142857142858	0.06998082470265705	1.1550209437950567
587	5	174	3	12	0.11494252873563218	0.016699929087495975	1.7772853729826312
622	6	173	3	12	0.13872832369942195	0.023865412628667555	1.6222310523597865
627	8	171	5	10	0.0935672514619883	0.001234261100135449	2.9085929583309174
639	9	170	5	10	0.10588235294117647	0.0018331517060542683	2.7368015926613842

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
640	10	169	5	10	0.11834319526627218	0.0026250181738802633	2.5808676854814183
657	11	168	5	10	0.13095238095238096	0.003644511061609513	2.438360727287861
663	13	166	5	10	0.1566265060240964	0.006510395780065948	2.1863926089934114
671	15	164	5	10	0.18292682926829268	0.01072245099455107	1.9697059299659876
673	16	163	5	10	0.19631901840490798	0.01342338349072885	1.8721380023380016
687	17	162	5	10	0.20987654320987653	0.01656697136370287	1.7807568784395744
708	18	161	5	10	0.2236024844720497	0.020185752752359003	1.694955050558838
712	88	91	8	7	0.8461538461538461	0.7940441928844832	0.10015532604674418
713	176	3	14	1	4.190476190476191	0.27712145920070164	0.5573298428514589
934	176	3	15	0	0.0	1.0	-0.0
1019	177	2	15	0	0.0	1.0	-0.0

2.3 domain_retention frame_domain_contingency_table

141	38
5	10

2.4 domain_retention domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained
kinase	Protein kinase domain	194	179	1	14	0.04988179669030733

2.5 frequency Partner_gene_counts

Partner_gene	Event_count
CCDC186	1
CCDC6	51
CLIP1	1
CTNNA1	1
CTNNA3	1
CUBN	1
DOCK1	1
EML4	1
ERC1	1
ERCC6	3
FBXL7	2
GEMIN5	1
GRIPAP1	1
KCNMA1	1
KIAA1217	1
KIF13A	1
KIF5B	87
LIN52	1
NCOA4	16
PDCD10	1
RASSF4	1
RBPMS	1
RELCH	1
RUFY1	2

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	is_main_ref	is_in_frame	is_retained	is_disrupted	description	annotation	effect_on_fra
EVT-P-0039548-T01-IM6-14	P-0039548-T01-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43610843	11	712	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-9501:CCDC6_c.2136+659:RETinv	NA	
EVT-P-0040111-T01-IM6-15	P-0040111-T01-IM6		CCDC6-RET	CCDC6	unknown	three_prime	43609967	11	640	False	retained	1.0	False	Protein kinase domain			Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-13873:CCDC6_c.1919:RETinv	NA	
EVT-P-0041192-T01-IM6-16	P-0041192-T01-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43610571	11	712	True	retained	1.0	False	Protein kinase domain			Fusion: mid-exon {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+8632:CCDC6_c.2136+387:RETinv	NA	
EVT-P-0022642-T04-IM6-17	P-0022642-T04-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43611768	12	713	True	retained	1.0	False	Protein kinase domain			Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-14213:CCDC6_c.2137-264:RETinv	NA	
EVT-P-0022642-T03-IM6-18	P-0022642-T03-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43611768	12	713	True	retained	1.0	False	Protein kinase domain			Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-14213:CCDC6_c.2137-264:RETinv	NA	
EVT-P-0051781-T01-IM6-19	P-0051781-T01-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43610377	11	712	True	retained	1.0	False	Protein kinase domain			Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+14702:CCDC6_c.2136+193:RETinv	NA	
EVT-P-0042417-T01-IM6-20	P-0042417-T01-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43610397	11	712	True	retained	1.0	False	Protein kinase domain			Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+16388:CCDC6_c.2136+213:RETinv	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_pos	protein_neg	domain_pos	domain_neg	main_retained	main_deleted	disrupted_domains	annotation	effect_on_fra	
EVT-P-0004901-T01-IM5-21	P-0004901-T01-IM5		CCDC6-RET	CCDC6	in-frame	three_prime	43611729	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) reciprocal fusion (CCDC6 exon1 with RET exons 12-20) : CCDC6_c. RETinv Protein fusion: in frame (CCDC6-RET)	NA (NM_005436) - RET (NM_020975)	
EVT-P-0022380-T03-IM6-22	P-0022380-T03-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43610659	11	712	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+21810:CCDC6_c.2136+475:RETinv Protein Fusion: in frame {CCDC6:RET}	NA (NM_005436) - RET (NM_020975)	2136+475:RETinv
EVT-P-0068500-T01-IM7-23	P-0068500-T01-IM7		CCDC6-RET	CCDC6	in-frame	three_prime	43610322	11	712	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+12590:CCDC6_c.2136+138:RETinv Protein Fusion: in frame {CCDC6:RET}	NA (NM_005436) - RET (NM_020975)	2136+138:RETinv
EVT-P-0057735-T01-IM6-24	P-0057735-T01-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43611887	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+125:CCDC6_c.2137-145:RETinv Protein Fusion: in frame {CCDC6:RET}	NA (NM_005436) - RET (NM_020975)	2137-145:RETinv
EVT-P-0038053-T01-IM6-26	P-0038053-T01-IM6		CLIP1-RET	CLIP1	unknown	three_prime	43611121	12	713	True	retained	1.0	False	Protein kinase domain			CLIP1 (NM_001247997) - RET (NM_020975) rearrangement: t(10;12)(q11.2;q24.31) Protein Fusion: in frame {CLIP1:RET}	NA (NM_001247997) - RET (NM_020975)	chr10:g.43611121
EVT-P-0000252-T01-IM3-27	P-0000252-T01-IM3		CUBN-RET	CUBN	unknown	three_prime	43610494	11	712	True	retained	1.0	False	Protein kinase domain			NA Antisense fusion	NA	
EVT-P-0038256-T01-IM6-28	P-0038256-T01-IM6		ERC1-RET	ERC1	unknown	three_prime	43609964	11	639	False	retained	1.0	False	Protein kinase domain			ERC1 (NM_178040) - RET (NM_020975) Rearrangement : t(10;12)(q11.1;p13.32)(chr10:g.43609964::chr12:g.126411111) Protein Fusion: mid-exon {ERC1:RET}	NA (NM_178040) - RET (NM_020975)	4::chr12:g.126411111

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_retained	protein_retained	breakpoint_4	domain_ref	domain_ref	domain_ref	domain_ref	domain_ref	disrupted_domain	description_annotation	effect_on_fusion
EVT-P-0009514-T01-IM5-34	P-0009514-T01-IM5		KIF5B-RET	KIF5B	in-frame	three_prime	43611811	12	713	True	retained	1.0	False	Protein kinase domain					Note: The KIF5B (NM_004521) - RET (NM_020975) reciprocal rearrangement event is an inversion which results in the fusion of KIF5B exons 1 to 15 and RET exons 12 to 20. The clinical research is currently ongoing with cabozantinib. PMID: 23533264 Protein fusion: in frame (KIF5B-RET) KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1798:KIF5B_c.2136+124:RETinv	NA
EVT-P-0055183-T01-IM6-35	P-0055183-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610308	11	712	True	retained	1.0	False	Protein kinase domain					KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1798:KIF5B_c.2136+124:RETinv	NA
EVT-P-0011035-T01-IM5-36	P-0011035-T01-IM5		KIF5B-RET	KIF5B	in-frame	three_prime	43611322	12	713	True	retained	1.0	False	Protein kinase domain					Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12-20) : c.1726-2296:KIF5B_c.2137-710:RETinv Protein fusion: in frame (KIF5B-RET) KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1863:KIF5B_c.2137-335:RETinv Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-0012648-T01-IM5-37	P-0012648-T01-IM5		KIF5B-RET	KIF5B	in-frame	three_prime	43611697	12	713	True	retained	1.0	False	Protein kinase domain					KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1863:KIF5B_c.2137-335:RETinv Protein fusion: in frame (KIF5B-RET)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	domain_ref	domain_1	domain_2	domain_3	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0014808-T02-IM6-44	P-0014808-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611056	11	712	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1765:KIF5B_c.2136+871:RETinv	NA	
EVT-P-0033700-T02-IM6-45	P-0033700-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611670	12	713	True	retained	1.0	False	Protein kinase domain				Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1454:KIF5B_c.2137-362:RET Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1764:KIF5B_c.2136+872:RETinv	NA	
EVT-P-0014808-T03-IM6-46	P-0014808-T03-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611056	11	712	True	retained	1.0	False	Protein kinase domain				Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1764:KIF5B_c.2136+872:RETinv	NA	
EVT-P-0008289-T01-IM5-47	P-0008289-T01-IM5		KIF5B-RET	KIF5B	in-frame	three_prime	43611748	12	713	True	retained	1.0	False	Protein kinase domain				Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20) : c.1726-824:KIF5B_c.2137-284:RETinv Protein fusion: in frame (KIF5B-RET) RET (NM_020975) Inversion (c.2136+418_KIF5B:c.1726-153inv) Protein fusion: in frame (KIF5B-RET) KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-18 fused with RET exons 12-20) : c.2095-38:KIF5B_c.2136+271:RETinv	NA	
EVT-P-0000252-T01-IM3-48	P-0000252-T01-IM3		KIF5B-RET	KIF5B	in-frame	three_prime	43610597	11	712	True	retained	1.0	False	Protein kinase domain				RET (NM_020975) Inversion (c.2136+418_KIF5B:c.1726-153inv) Protein fusion: in frame (KIF5B-RET) KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-18 fused with RET exons 12-20) : c.2095-38:KIF5B_c.2136+271:RETinv	NA	
EVT-P-0008101-T01-IM5-49	P-0008101-T01-IM5		KIF5B-RET	KIF5B	in-frame	three_prime	43610455	11	712	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-18 fused with RET exons 12-20) : c.2095-38:KIF5B_c.2136+271:RETinv	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	clonally_expressed	in_frame	domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0048968-T01-IM6-50	P-0048968-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611809	12	713	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion: c.1914+211:KIF5B_c.2137-223:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B NA (NM_004521) - RET (NM_020975) fusion: c.1725+948:KIF5B_c.2136+432:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B NA (NM_004521) - RET (NM_020975) fusion: c.1726-1902:KIF5B_c.2137-733:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B NA (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused in-frame to RET exons 12-20) : c.1725+275:KIF5B_c.2137-764:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B NA (NM_004521) - RET (NM_020975) fusion: c.1725+1038:KIF5B_c.2137-158:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B NA (NM_004521) - RET (NM_020975) fusion: c.1725+2508:KIF5B_c.2136+922:RETinv Protein Fusion: in frame {KIF5B:RET}	NA	
EVT-P-0048829-T01-IM6-51	P-0048829-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610616	11	712	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+948:KIF5B_c.2136+432:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B NA (NM_004521) - RET (NM_020975) fusion: c.1726-1902:KIF5B_c.2137-733:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B NA (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused in-frame to RET exons 12-20) : c.1725+275:KIF5B_c.2137-764:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B NA (NM_004521) - RET (NM_020975) fusion: c.1725+1038:KIF5B_c.2137-158:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B NA (NM_004521) - RET (NM_020975) fusion: c.1725+2508:KIF5B_c.2136+922:RETinv Protein Fusion: in frame {KIF5B:RET}	NA	
EVT-P-0065876-T01-IM7-52	P-0065876-T01-IM7		KIF5B-RET	KIF5B	in-frame	three_prime	43611299	12	713	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-1902:KIF5B_c.2137-733:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B NA (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused in-frame to RET exons 12-20) : c.1725+275:KIF5B_c.2137-764:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B NA (NM_004521) - RET (NM_020975) fusion: c.1725+1038:KIF5B_c.2137-158:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B NA (NM_004521) - RET (NM_020975) fusion: c.1725+2508:KIF5B_c.2136+922:RETinv Protein Fusion: in frame {KIF5B:RET}	NA	
EVT-P-0016221-T01-IM6-53	P-0016221-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611268	12	713	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-1902:KIF5B_c.2137-733:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B NA (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused in-frame to RET exons 12-20) : c.1725+275:KIF5B_c.2137-764:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B NA (NM_004521) - RET (NM_020975) fusion: c.1725+1038:KIF5B_c.2137-158:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B NA (NM_004521) - RET (NM_020975) fusion: c.1725+2508:KIF5B_c.2136+922:RETinv Protein Fusion: in frame {KIF5B:RET}	NA	
EVT-P-0037367-T01-IM6-54	P-0037367-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611874	12	713	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1038:KIF5B_c.2137-158:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B NA (NM_004521) - RET (NM_020975) fusion: c.1725+2508:KIF5B_c.2136+922:RETinv Protein Fusion: in frame {KIF5B:RET}	NA	
EVT-P-0057072-T02-IM6-55	P-0057072-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611106	11	712	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2508:KIF5B_c.2136+922:RETinv Protein Fusion: in frame {KIF5B:RET}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1_pos	protein_2_pos	breakpoint_4	main_ref	inframe	inframe	inframe	inframe	disrupted_domains	description	annotation	effect_on_fra
EVT-P-0046827-T02-IM6-62	P-0046827-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610889	11	712	True	retained	1.0	False	Protein kinase domain					KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-1305:KIF5B_c.2136+705:RETinv		
EVT-P-0032407-T01-IM6-63	P-0032407-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611730	12	713	True	retained	1.0	False	Protein kinase domain					Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.1914+230:KIF5B_c.2137-302:RETinv		
EVT-P-0018069-T01-IM6-64	P-0018069-T01-IM6		KIF5B-RET	KIF5B	out-of-frame	five_prime	43607832	8	550	True	lost	0.0	False		Protein kinase domain				Fusion: in frame {KIF5B:RET} RET (NM_020975) - KIF5B (NM_004521) rearrangement: c.1648+160:RET_c.1915-72:KIF5Binv		
EVT-P-0005573-T01-IM5-65	P-0005573-T01-IM5		KIF5B-RET	KIF5B	in-frame	three_prime	43610418	11	712	True	retained	1.0	False	Protein kinase domain					Fusion: out of frame {RET:KIF5B} KIF5B (NM_004521) - RET (NM_020975) Fusion (KIF5B exon15 fused with RET exon12): c.1726-860:KIF5B_c.2136+234:RETinv		
EVT-P-0004995-T01-IM5-66	P-0004995-T01-IM5		KIF5B-RET	KIF5B	in-frame	three_prime	43610660	11	712	True	retained	1.0	False	Protein kinase domain					Fusion: in frame (KIF5B-RET) KIF5B (NM_004521) -RET (NM_020975) Fusion (KIF5B exon 15 fused to RET exon 12) : c.1726-2366:KIF5B_c.2136+478:RETinv		
EVT-P-0036092-T01-IM6-67	P-0036092-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610413	11	712	True	retained	1.0	False	Protein kinase domain					Fusion: in frame (KIF5B-RET) KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1138:KIF5B_c.2136+229:RETinv		

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	donor_frame	inframe	domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0046726-T01-IM6-68	P-0046726-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610987	11	712	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+768:KIF5B_c.2136+803:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2000:KIF5B_c.2137-907:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1914+131:KIF5B_c.2137-213:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.2544+168:KIF5B_c.2136+629:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+312:KIF5B_c.2137-441:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1231:KIF5B_c.2136+624:RETinv Protein Fusion: in frame {KIF5B:RET}	NA	
EVT-P-0032227-T01-IM6-69	P-0032227-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611125	12	713	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2000:KIF5B_c.2137-907:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1914+131:KIF5B_c.2137-213:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.2544+168:KIF5B_c.2136+629:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+312:KIF5B_c.2137-441:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1231:KIF5B_c.2136+624:RETinv Protein Fusion: in frame {KIF5B:RET}	NA	
EVT-P-0020273-T02-IM6-70	P-0020273-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611819	12	713	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1914+131:KIF5B_c.2137-213:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.2544+168:KIF5B_c.2136+629:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+312:KIF5B_c.2137-441:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1231:KIF5B_c.2136+624:RETinv Protein Fusion: in frame {KIF5B:RET}	NA	
EVT-P-0043007-T02-IM6-71	P-0043007-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610813	11	712	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion: c.2544+168:KIF5B_c.2136+629:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+312:KIF5B_c.2137-441:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1231:KIF5B_c.2136+624:RETinv Protein Fusion: in frame {KIF5B:RET}	NA	
EVT-P-0021150-T01-IM6-72	P-0021150-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611591	12	713	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+312:KIF5B_c.2137-441:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1231:KIF5B_c.2136+624:RETinv Protein Fusion: in frame {KIF5B:RET}	NA	
EVT-P-0021578-T01-IM6-73	P-0021578-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610809	11	712	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+312:KIF5B_c.2137-441:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1231:KIF5B_c.2136+624:RETinv Protein Fusion: in frame {KIF5B:RET}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	donor_frame	inframe	domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0039941-T01-IM6-93	P-0039941-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611339	12	713	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2040:KIF5B_c.2137-693:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B NA (NM_004521) - RET (NM_020975) fusion: c.1726-2304:KIF5B_c.2137-505:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B NA (NM_004521) - RET (NM_020975) fusion: c.1915-203:KIF5B_c.2136+175:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B NA (NM_004521) - RET (NM_020975) fusion: c.1725+1798:KIF5B_c.2136+124:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B NA (NM_004521) - RET (NM_020975) fusion: (KIF5B exons 1-15 fused with RET exons 12-20.): c.1726-601:KIF5B_c.2136+788:RETinv Protein Fusion: in frame {KIF5B:RET}	NA NA c.2137- c.2137- c.2136+175:RETinv c.2136+175:RETinv c.2136+124:RETinv c.2136+788:RETinv	
EVT-P-0060748-T03-IM7-94	P-0060748-T03-IM7		KIF5B-RET	KIF5B	in-frame	three_prime	43611527	12	713	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2304:KIF5B_c.2137-505:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B NA (NM_004521) - RET (NM_020975) fusion: c.1915-203:KIF5B_c.2136+175:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B NA (NM_004521) - RET (NM_020975) fusion: c.1725+1798:KIF5B_c.2136+124:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B NA (NM_004521) - RET (NM_020975) fusion: (KIF5B exons 1-15 fused with RET exons 12-20.): c.1726-601:KIF5B_c.2136+788:RETinv Protein Fusion: in frame {KIF5B:RET}	NA NA c.2137- c.2137- c.2136+175:RETinv c.2136+175:RETinv c.2136+124:RETinv c.2136+788:RETinv	
EVT-P-0039474-T02-IM6-95	P-0039474-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610359	11	712	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion: c.1915-203:KIF5B_c.2136+175:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B NA (NM_004521) - RET (NM_020975) fusion: c.1725+1798:KIF5B_c.2136+124:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B NA (NM_004521) - RET (NM_020975) fusion: (KIF5B exons 1-15 fused with RET exons 12-20.): c.1726-601:KIF5B_c.2136+788:RETinv Protein Fusion: in frame {KIF5B:RET}	NA NA c.2136+175:RETinv c.2136+175:RETinv c.2136+124:RETinv c.2136+788:RETinv	
EVT-P-0039474-T01-IM6-96	P-0039474-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610359	11	712	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion: c.1915-203:KIF5B_c.2136+175:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B NA (NM_004521) - RET (NM_020975) fusion: c.1725+1798:KIF5B_c.2136+124:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B NA (NM_004521) - RET (NM_020975) fusion: (KIF5B exons 1-15 fused with RET exons 12-20.): c.1726-601:KIF5B_c.2136+788:RETinv Protein Fusion: in frame {KIF5B:RET}	NA NA c.2136+175:RETinv c.2136+175:RETinv c.2136+124:RETinv c.2136+788:RETinv	
EVT-P-0055183-T02-IM6-97	P-0055183-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610308	11	712	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1798:KIF5B_c.2136+124:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B NA (NM_004521) - RET (NM_020975) fusion: (KIF5B exons 1-15 fused with RET exons 12-20.): c.1726-601:KIF5B_c.2136+788:RETinv Protein Fusion: in frame {KIF5B:RET}	NA NA c.2136+124:RETinv c.2136+788:RETinv	
EVT-P-0025838-T01-IM6-98	P-0025838-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610972	11	712	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion: (KIF5B exons 1-15 fused with RET exons 12-20.): c.1726-601:KIF5B_c.2136+788:RETinv Protein Fusion: in frame {KIF5B:RET}	NA NA c.2136+788:RETinv	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_coding	breakpoint_4	breakpoint_5	main_ref	donor_frame	maintained	disrupted_domain	disrupted_domain	description_annotation	status	effect_on_fra
EVT-P-0025928-T01-IM6-99	P-0025928-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611910	12	713	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+483:KIF5B_c.2137-122C>T:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+483:KIF5B_c.2137-122:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-1731:KIF5B_c.2137-67:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1132:KIF5B_c.2012:RETinv Protein Fusion: mid-exon {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.2544+156:KIF5_c.2136+327:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1726-1554:KIF5B_c.2136+538:RETinv Protein Fusion: in frame {KIF5B:RET}	NA	
EVT-P-0025928-T02-IM6-100	P-0025928-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611910	12	713	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+483:KIF5B_c.2137-122:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-1731:KIF5B_c.2137-67:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.2544+156:KIF5_c.2136+327:RETinv Protein Fusion: mid-exon {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-1554:KIF5B_c.2136+538:RETinv Protein Fusion: in frame {KIF5B:RET}	NA	
EVT-P-0036637-T01-IM6-101	P-0036637-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611965	12	713	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-1731:KIF5B_c.2137-67:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1132:KIF5B_c.2012:RETinv Protein Fusion: mid-exon {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.2544+156:KIF5_c.2136+327:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1726-1554:KIF5B_c.2136+538:RETinv Protein Fusion: in frame {KIF5B:RET}	NA	
EVT-P-0039257-T01-IM6-102	P-0039257-T01-IM6		KIF5B-RET	KIF5B	unknown	three_prime	43610060	11	671	False	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1132:KIF5B_c.2012:RETinv Protein Fusion: mid-exon {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.2544+156:KIF5_c.2136+327:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1726-1554:KIF5B_c.2136+538:RETinv Protein Fusion: in frame {KIF5B:RET}	NA	
EVT-P-0026693-T01-IM6-103	P-0026693-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610511	11	712	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion: c.2544+156:KIF5_c.2136+327:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1726-1554:KIF5B_c.2136+538:RETinv Protein Fusion: in frame {KIF5B:RET}	NA	
EVT-P-0026777-T01-IM6-104	P-0026777-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610717	11	712	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1726-1554:KIF5B_c.2136+538:RETinv Protein Fusion: in frame {KIF5B:RET}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	donor_frame	inframe	disrupted_domain	disrupted_domain	description_annotation	effect_on_fra
EVT-P-0026777-T02-IM6-105	P-0026777-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610717	11	712	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1726-1554:KIF5B_c.2136+538:RETinv	
EVT-P-0038961-T01-IM6-106	P-0038961-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610775	11	712	True	retained	1.0	False	Protein kinase domain				Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-293:KIF5B_c.2136+591:RETinv	
EVT-P-0027723-T01-IM6-107	P-0027723-T01-IM6		KIF5B-RET	KIF5B	unknown	three_prime	43610171	11	708	False	retained	1.0	False	Protein kinase domain				Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 11-20):c.1725+966:KIF5B_c.2123:RETinv	
EVT-P-0036737-T02-IM6-108	P-0036737-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43612025	12	713	True	retained	1.0	False	Protein kinase domain				mid-exon {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1487:RETinv	
EVT-P-0061331-T01-IM7-109	P-0061331-T01-IM7		KIF5B-RET	KIF5B	in-frame	three_prime	43610884	11	712	True	retained	1.0	False	Protein kinase domain				Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.1915-286:KIF5B_c.2136+700:RETinv	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_coding	breakpoint_4	breakpoint_5	main_ref	donor_frame	inframe	disrupted_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0027991-T01-IM6-110	P-0027991-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43608201	9	550	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-24 fused to RET exons 9-20): c.2762-248:KIF5B_c.1649-100:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1625:KIF5B_c.2136+27:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.2204+2:KIF5B_c.2137-902:RETinv Protein Fusion: out of frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2396:KIF5B_c.2136+727:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exon15 fused with RET exon12): c.1725+985:KIF5B_c.2137-277:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12- 20): c.1726-1971:KIF5B_c.2137-385:RETinv Protein fusion: in frame (KIF5B-RET)	NA	
EVT-P-0062948-T03-IM7-111	P-0062948-T03-IM7		KIF5B-RET	KIF5B	in-frame	three_prime	43610211	11	712	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1625:KIF5B_c.2136+27:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.2204+2:KIF5B_c.2137-902:RETinv Protein Fusion: out of frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2396:KIF5B_c.2136+727:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exon15 fused with RET exon12): c.1725+985:KIF5B_c.2137-277:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12- 20): c.1726-1971:KIF5B_c.2137-385:RETinv Protein fusion: in frame (KIF5B-RET)	NA	
EVT-P-0064951-T01-IM7-112	P-0064951-T01-IM7		KIF5B-RET	KIF5B	out-of-frame	three_prime	43611130	12	713	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion: c.2204+2:KIF5B_c.2137-902:RETinv Protein Fusion: out of frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2396:KIF5B_c.2136+727:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exon15 fused with RET exon12): c.1725+985:KIF5B_c.2137-277:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12- 20): c.1726-1971:KIF5B_c.2137-385:RETinv Protein fusion: in frame (KIF5B-RET)	NA	
EVT-P-0030453-T01-IM6-113	P-0030453-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610911	11	712	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2396:KIF5B_c.2136+727:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exon15 fused with RET exon12): c.1725+985:KIF5B_c.2137-277:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12- 20): c.1726-1971:KIF5B_c.2137-385:RETinv Protein fusion: in frame (KIF5B-RET)	NA	
EVT-P-0037494-T01-IM6-114	P-0037494-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611755	12	713	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exon15 fused with RET exon12): c.1725+985:KIF5B_c.2137-277:RETinv Protein Fusion: in frame {KIF5B:RET} KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12- 20): c.1726-1971:KIF5B_c.2137-385:RETinv Protein fusion: in frame (KIF5B-RET)	NA	
EVT-P-0000208-T02-IM5-115	P-0000208-T02-IM5		KIF5B-RET	KIF5B	in-frame	three_prime	43611647	12	713	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12- 20): c.1726-1971:KIF5B_c.2137-385:RETinv Protein fusion: in frame (KIF5B-RET)	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	donor_frame	inframe	disrupted_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0052409-T01-IM6-130	P-0052409-T01-IM6		NCOA4-RET	NCOA4	out-of-frame	three_prime	43611691	12	713	True	retained	1.0	False	Protein kinase domain				NCOA4 (NM_001145260) - RET (NM_020975) rearrangement: c.618+214:NCOA4_c.2137-341:RETdup Protein Fusion: out of frame {TIMM23B:RET} KIAA1468 (NM_020854) - RET (NM_020975) fusion (KIAA1468 exons 1-10 fused with RET exons 12-20): t(10;18)(q11.2;q21.33)(chr10:g.4361027	NA 5260) RET (NM_020975) COA4_c.2137-	
EVT-P-0004094-T02-IM6-131	P-0004094-T02-IM6		RELCH-RET	RELCH	in-frame	three_prime	43610279	11	712	True	retained	1.0	False	Protein kinase domain				fusion (KIAA1468 exons 1-10 fused with RET exons 12-20): t(10;18)(q11.2;q21.33)(chr10:g.4361027	RET} NA (NM_020854) RET (NM_020975) fusion (KIAA1468 exons 1-10 fused with RET exons 12-20): t(10;18)(q11.2;q21.33)(chr10:g.4361027	
EVT-P-0015588-T01-IM6-132	P-0015588-T01-IM6		RET-CCDC186	CCDC186	in-frame	three_prime	43610308	11	712	True	retained	1.0	False	Protein kinase domain				fusion (KIAA1468 exons 1-10 fused with RET exons 12-20): t(10;18)(q11.2;q21.33)(chr10:g.4361027	RET} NA (NM_018017) RET (NM_020975) Rearrangement: c.1326+142:C10orf118_c.2136+124:RET	
EVT-P-0020974-T01-IM6-133	P-0020974-T01-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611184	12	713	True	retained	1.0	False	Protein kinase domain				fusion (CCDC6 exon 1 fused to RET exon 12): c.303+2543:848:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - (RET (NM_020975) rearrangement: c.304-9458:CCDC6_c.2137-787:RETinv Protein fusion: in frame (CCDC6-RET)	NA (NM_005436) RET (NM_020975) CCDC6_c.2137-	
EVT-P-0001745-T02-IM5-134	P-0001745-T02-IM5		RET-CCDC6	CCDC6	in-frame	three_prime	43611245	12	713	True	retained	1.0	False	Protein kinase domain				fusion (CCDC6 exon 1 fused to RET exon 12): c.303+2543:848:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - (RET (NM_020975) rearrangement: c.304-9458:CCDC6_c.2137-787:RETinv Protein fusion: in frame (CCDC6-RET)	NA (NM_005436) RET (NM_020975) CCDC6_c.2137-	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_pos	domain_pos	main_retained	fusion_retained	domain_pos	disrupted_domains	domains_annotation	effect_on_fusion
EVT-P-0017365-T02-IM6-135	P-0017365-T02-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611464	12	713	True	retained	1.0	False	Protein kinase domain		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+25501:CCDC6_c.2137-568:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 - RET (NM_005436) - RET (NM_020975) inversion: c.304-12339_c.2137-333inv5 Protein fusion: in frame (CCDC6-RET) CCDC6 - RET (NM_005436) - RET (NM_020975) fusion: c.304-5768:CCDC6_c.2137-872:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436)-RET (NM_020975) Fusion (CCDC6 exon 2 fused to RET exon11): c.304-1153:CCDC6_c.2136+169:RETinv	2137-
EVT-P-0001898-T01-IM3-136	P-0001898-T01-IM3		RET-CCDC6	CCDC6	in-frame	three_prime	43611699	12	713	True	retained	1.0	False	Protein kinase domain		CCDC6 (NM_005436) - RET (NM_020975) inversion: c.304-12339_c.2137-333inv5 Protein fusion: in frame (CCDC6-RET) CCDC6 - RET (NM_005436) - RET (NM_020975) fusion: c.304-5768:CCDC6_c.2137-872:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436)-RET (NM_020975) Fusion (CCDC6 exon 2 fused to RET exon11): c.304-1153:CCDC6_c.2136+169:RETinv	2137-
EVT-P-0029602-T01-IM6-137	P-0029602-T01-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611160	12	713	True	retained	1.0	False	Protein kinase domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-5768:CCDC6_c.2137-872:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436)-RET (NM_020975) Fusion (CCDC6 exon 2 fused to RET exon11): c.304-1153:CCDC6_c.2136+169:RETinv	2137-
EVT-P-0004916-T01-IM5-138	P-0004916-T01-IM5		RET-CCDC6	CCDC6	in-frame	three_prime	43610353	11	712	True	retained	1.0	False	Protein kinase domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-1153:CCDC6_c.2136+169:RETinv	2136+169:RETinv
EVT-P-0000425-T01-IM3-139	P-0000425-T01-IM3		RET-CCDC6	CCDC6	in-frame	five_prime	43610450	11	712	True	lost	0.0	False	Protein kinase domain		RET (NM_020630) - CCDC6 (NM_005436) Inversion (c.2136+266_c.303+19630inv) Protein fusion: in frame (RET-CCDC6) CCDC6 - RET (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+3703:CCDC6_c.2136+169:RETinv	NA
EVT-P-0031330-T01-IM6-140	P-0031330-T01-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43610353	11	712	True	retained	1.0	False	Protein kinase domain		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+3703:CCDC6_c.2136+169:RETinv	2136+169:RETinv

[illegible]

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1_pos	protein_2_pos	breakpoint_4	is_main_ref	is_in_frame	is_retained	is_lost	disrupted_domains	description	annotation	effect_on_fra
EVT-P-0029315-T01-IM6-146	P-0029315-T01-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43610631	11	712	True	retained	1.0	False	Protein kinase domain				CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.304-20462:CCDC6_c.2136+447:RETinv	NA	
EVT-P-0000889-T01-IM3-147	P-0000889-T01-IM3		RET-CCDC6	CCDC6	in-frame	five_prime	43611994	12	713	True	lost	0.0	False	Protein kinase domain				RET (NM_020630) - CCDC6 (NM_005436) Reciprocal Inversion: c.2137-38_c.2137-38inv Protein fusion: in frame {CCDC6:RET} RET (NM_020630) - CCDC6 (NM_005436) Reciprocal Inversion: c.2137-38_c.2137-38inv Protein fusion: in frame (RET-CCDC6) CCDC6 (NM_005436) - RET (NM_020975) fusion: c.454-1512:CCDC6_c.2136+84:RETinv	NA	
EVT-P-0007322-T02-IM6-148	P-0007322-T02-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43610268	11	712	True	retained	1.0	False	Protein kinase domain				CCDC6 (NM_005436) - RET (NM_020975) fusion: c.454-1512:CCDC6_c.2136+84:RETinv	NA	
EVT-P-0027953-T01-IM6-149	P-0027953-T01-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611278	12	713	True	retained	1.0	False	Protein kinase domain				CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+20901:CCDC6_c.2137-754:RETinv	NA	
EVT-P-0008504-T03-IM6-150	P-0008504-T03-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43610583	11	712	True	retained	1.0	False	Protein kinase domain				CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exons 1-8 fused to RET exons 12-20): c.1230+499:CCDC6_c.2136+399:RETinv	NA	
EVT-P-0008955-T03-IM5-151	P-0008955-T03-IM5		RET-CCDC6	CCDC6	in-frame	three_prime	43611019	11	712	True	retained	1.0	False	Protein kinase domain				CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+26028:CCDC6_c.2136+835:RETinv	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	is_main_ref	is_in_frame	is_retained	is_disrupted	description	annotation	effect_on_fra
EVT-P-0013023-T01-IM5-152	P-0013023-T01-IM5		RET-CCDC6	CCDC6	in-frame	three_prime	43611810	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-17904:CCDC6_c.2137-222:RETinv Protein fusion: in frame (CCDC6-RET) CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+18884:CCDC6_c.2136+48:RETinv Protein fusion: in frame (CCDC6-RET) CCDC6 (NM_005436) - RET (NM_020975) fusion: exon 1 fused to RET exons 12-20): c.304-414:CCDC6_c.2137-820:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6(NM_005436) - RET(NM_020975) Fusion (CCDC6 exon1 fused to RET exon12) : c.304-14214:CCDC6_c.2137-273:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) Fusion (CCDC6 exon1 fused to RET exons 12-20) : c.304-14214:CCDC6_c.2137-273:RETinv Protein Fusion: in frame {CCDC6:RET}	NA	
EVT-P-0013394-T01-IM5-153	P-0013394-T01-IM5		RET-CCDC6	CCDC6	in-frame	three_prime	43610232	11	712	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+18884:CCDC6_c.2136+48:RETinv Protein fusion: in frame (CCDC6-RET) CCDC6 (NM_005436) - RET (NM_020975) fusion: exon 1 fused to RET exons 12-20): c.304-414:CCDC6_c.2137-820:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6(NM_005436) - RET(NM_020975) Fusion (CCDC6 exon1 fused to RET exon12) : c.304-14214:CCDC6_c.2137-273:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) Fusion (CCDC6 exon1 fused to RET exons 12-20) : c.304-14214:CCDC6_c.2137-273:RETinv Protein Fusion: in frame {CCDC6:RET}	NA	
EVT-P-0022607-T01-IM6-154	P-0022607-T01-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611212	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: exon 1 fused to RET exons 12-20): c.304-414:CCDC6_c.2137-820:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6(NM_005436) - RET(NM_020975) Fusion (CCDC6 exon1 fused to RET exon12) : c.304-14214:CCDC6_c.2137-273:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) Fusion (CCDC6 exon1 fused to RET exons 12-20) : c.304-14214:CCDC6_c.2137-273:RETinv Protein Fusion: in frame {CCDC6:RET}	NA	
EVT-P-0022642-T01-IM6-155	P-0022642-T01-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611768	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: exon 1 fused to RET exon12) : c.304-14214:CCDC6_c.2137-273:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6(NM_005436) - RET(NM_020975) Fusion (CCDC6 exon1 fused to RET exon12) : c.304-14214:CCDC6_c.2137-273:RETinv Protein Fusion: in frame {CCDC6:RET}	NA	
EVT-P-0022642-T02-IM6-156	P-0022642-T02-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611768	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: exon 1 fused to RET exon12) : c.304-14214:CCDC6_c.2137-273:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6(NM_005436) - RET(NM_020975) Fusion (CCDC6 exon1 fused to RET exon12) : c.304-14214:CCDC6_c.2137-273:RETinv Protein Fusion: in frame {CCDC6:RET}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	is_main_ref	is_in_frame	is_in_domain	is_disrupted	description	annotation	effect_on_fra
EVT-P-0017365-T03-IM6-157	P-0017365-T03-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611464	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+25501:568:RETinv Protein Fusion: in frame {CCDC6:RET} CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.304-10038:CCDC6_c.2137-46:RETinv Protein Fusion: in frame {CCDC6:RET} RET (NM_020975) - CCDC6 (NM_005436) rearrangement: c.1879+314:RET_c.304-15222:CCDC6inv Protein Fusion: out of frame {RET:CCDC6} CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exon 12): c.304-21366:CCDC6_c.2136+785:RETinv	NA	
EVT-P-0025349-T03-IM6-158	P-0025349-T03-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611986	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.304-10038:CCDC6_c.2137-46:RETinv Protein Fusion: in frame {CCDC6:RET} RET (NM_020975) - CCDC6 (NM_005436) rearrangement: c.1879+314:RET_c.304-15222:CCDC6inv Protein Fusion: out of frame {RET:CCDC6} CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exon 12): c.304-21366:CCDC6_c.2136+785:RETinv	NA	
EVT-P-0049588-T01-IM6-159	P-0049588-T01-IM6		RET-CCDC6	CCDC6	out-of-frame	five_prime	43609437	10	627	True	lost	0.0	False		Protein kinase domain		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.304-10038:CCDC6_c.2137-46:RETinv Protein Fusion: in frame {CCDC6:RET} RET (NM_020975) - CCDC6 (NM_005436) rearrangement: c.1879+314:RET_c.304-15222:CCDC6inv Protein Fusion: out of frame {RET:CCDC6} CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exon 12): c.304-21366:CCDC6_c.2136+785:RETinv	NA	
EVT-P-0015435-T01-IM6-160	P-0015435-T01-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43610969	11	712	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exon 12): c.304-21366:CCDC6_c.2136+785:RETinv	NA	
EVT-P-0005556-T01-IM5-161	P-0005556-T01-IM5		RET-CCDC6	CCDC6	in-frame	three_prime	43611828	12	713	True	retained	1.0	False	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon1 with RET exons 12-20) - c.303+21888:CCDC6_c.2137-204:RETinv Protein fusion: in frame (CCDC6-RET)	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint	breakpoint	breakpoint	protein_pos	domain_pos	main_retained	dock_fusion	retained_def	lost_domain	disrupted_domain	deletion_annotation	station_site	effect_on_frame
EVT-P-0063502-T01-IM7-162	P-0063502-T01-IM7		RET-CTNNA1	CTNNA1	unknown	five_prime	43622038	19	1019	False	retained	1.0	False	Protein kinase domain			RET (NM_020975) - CTNNA1 (NM_001903) fusion: t(5;10)(q31.2;q11.21)(chr5:g.138223827: chr10:g.111711111)	NA	
EVT-P-0063802-T01-IM7-163	P-0063802-T01-IM7		RET-CTNNA3	CTNNA3	out-of-frame	five_prime	43617566	16	934	True	disrupted	0.74822695035461	True		Protein kinase domain		Fusion: mid-exon {RET:CTNNA1} RET (NM_020975) - CTNNA3 (NM_013266) rearrangement: c.2801+102:RET_c.1375-34298:CTNNA3inv Protein Fusion: out of frame {RET:CTNNA3}	NA	
EVT-P-0030396-T02-IM6-164	P-0030396-T02-IM6		RET-DOCK1	DOCK1	unknown	three_prime	43608320	9	556	False	retained	1.0	False	Protein kinase domain			Fusion: out of frame {RET:CTNNA3} DOCK1 (NM_001380) - RET (NM_020975) Rearrangement: c.2515+295:RET_c.1668:RETdup Protein Fusion: mid-exon {DOCK1:RET} EML4 (NM_019063) - RET (NM_020975) rearrangement: t(2;10)(p21;q11.21)(chr2:g.42550559::chr10:g.111711111)	NA	
EVT-P-0013886-T01-IM5-165	P-0013886-T01-IM5		RET-EML4	EML4	unknown	three_prime	43610067	11	673	False	retained	1.0	False	Protein kinase domain			Fusion: mid-exon (EML4-RET) RET (NM_020975) - ERCC6 (NM_000124) rearrangement: c.2137-816:RET_c.1397+6822:ERCC6inv Protein Fusion: out of frame {RET:ERCC6}	NA	
EVT-P-0014808-T03-IM6-166	P-0014808-T03-IM6		RET-ERCC6	ERCC6	out-of-frame	five_prime	43611216	12	713	True	lost	0.0	False		Protein kinase domain		Fusion: out of frame {RET:ERCC6} RET (NM_020975) - ERCC6 (NM_000124) rearrangement: c.2137-816:RET_c.1397+6822:ERCC6inv Protein Fusion: out of frame {RET:ERCC6}	NA	
EVT-P-0014808-T02-IM6-167	P-0014808-T02-IM6		RET-ERCC6	ERCC6	out-of-frame	five_prime	43611216	12	713	True	lost	0.0	False		Protein kinase domain		Fusion: out of frame {RET:ERCC6} RET (NM_020975) - ERCC6 (NM_000124) rearrangement: c.2137-816:RET_c.1397+6822:ERCC6inv Protein Fusion: out of frame {RET:ERCC6}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	chr1	start	end	domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0030396-T02-IM6-185	P-0030396-T02-IM6		RET-RBPMS	RBPMS	unknown	three_prime	43610018	11	657	False	retained	1.0	False	Protein kinase domain					RBPMS (NM_001008712) - RET (NM_020975) Rear-range-ment : t(8;10)(p21,q11.21)(chr8:g.30412656:::chr10:g.72412310::) Protein Fusion: mid-exon {RBPMS:RET} RUFY1 (NM_025158) - RET (NM_020975) Rear-range-ment : t(5;10)(q35.3,q11.21)(chr5:g.179025890:::chr11:g.71655251::)	NA	
EVT-P-0029222-T02-IM6-212	P-0029222-T02-IM6		RET-RUFY1	RUFY1	unknown	three_prime	43610037	11	663	False	retained	1.0	False	Protein kinase domain					RUFY1 (NM_001037442) - RET (NM_020975) Rear-range-ment : t(5;10)(q35.3,q11.21)(chr5:g.179025890:::chr11:g.71655251::)	NA	
EVT-P-0015310-T02-IM6-213	P-0015310-T02-IM6		RET-RUFY3	RUFY3	out-of-frame	three_prime	43610427	11	712	True	retained	1.0	False	Protein kinase domain					RUFY3 (NM_001037442) - RET (NM_020975) rearrange-ment: t(4;10)(q13.3;q11.21)(chr4:g.71655316:::chr11:g.71655251::)	NA	
EVT-P-0015310-T04-IM6-214	P-0015310-T04-IM6		RET-RUFY3	RUFY3	unknown	three_prime	43610427	11	712	True	retained	1.0	False	Protein kinase domain					RUFY3 (NM_001037442) - RET (NM_020975) rearrange-ment: t(4;10)(q13.3;q11.21)(chr4:g.71655251:::chr11:g.71655251::)	NA	
EVT-P-0025609-T01-IM6-215	P-0025609-T01-IM6		RET-SHROOM3	SHROOM3	in-frame	three_prime	43611545	12	713	True	retained	1.0	False	Protein kinase domain					SHROOM3 (NM_020859) - RET (NM_020975) rearrange-ment: t(4;10)(q21.1;q11.21)(chr4:g.77664882:::chr11:g.71655251::)	NA	
EVT-P-0053530-T01-IM6-216	P-0053530-T01-IM6		RET-SLC4A4	SLC4A4	out-of-frame	five_prime	43610439	11	712	True	lost	0.0	False	Protein kinase domain			Protein kinase domain		RET (NM_020975) - SLC4A4 (NM_001134742) rearrange-ment: t(4;10)(q13.3;q11.21)(chr4:g.72412310:::chr11:g.71655251::)	NA	
EVT-P-0002145-T01-IM3-217	P-0002145-T01-IM3		RET-TFG	TFG	in-frame	three_prime	43609437	10	627	True	retained	1.0	False	Protein kinase domain					TFG (NM_001195478) translo-cation: t(10;3)(q11.21;q12.2)(chr10:g.43609437:::chr3:g.171700000::)	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint	breakpoint	breakpoint	protein_pos	protein_pos	main_ret	cond	fract	unintended	host_domain	disrupted_domains	annotation	station_site	effect_on_fra
EVT-P-0021980-T01-IM6-218	P-0021980-T01-IM6		RET-TIMM23B	TIMM23B	out-of-frame	three_prime	43610550	11	712	True	retained	1.0	False	Protein kinase domain			NCOA4 (NM_001145260) - RET (NM_020975) fusion (NCOA4 exons 1-8 fused to RET exons 12-20): c.763-591:NCOA4_c.2136+366:RETdup			
EVT-P-0017716-T01-IM6-219	P-0017716-T01-IM6		RET-TIMM23B	TIMM23B	out-of-frame	three_prime	43611172	12	713	True	retained	1.0	False	Protein kinase domain			{TIMM23B:RET} NCOA4 (NM_001145260) - RET (NM_020975) fusion (NCOA4 exons 1-7 fused with RET exons 12-20): c.618+42:NCOA4_c.2137-860:RETdup			
EVT-P-0025531-T01-IM6-220	P-0025531-T01-IM6		RET-TIMM23B	TIMM23B	out-of-frame	three_prime	43611375	12	713	True	retained	1.0	False	Protein kinase domain			{TIMM23B:RET} NCOA4 (NM_001145260) RET (NM_020975) fusion: c.619-108NCOA4_c.2137-657RETdup			
EVT-P-0034704-T01-IM6-222	P-0034704-T01-IM6		RET-TRIM24	TRIM24	in-frame	three_prime	43611827	12	713	True	retained	1.0	False	Protein kinase domain			{TIMM23B:RET} TRIM24 (NM_015905) - RET (NM_020975) Fusion (TRIM24 exon 9 fused with RET exon12) : t(7;10)(q34;q11.21)(chr7:g.138243011::cl			
EVT-P-0046790-T01-IM6-223	P-0046790-T01-IM6		RET-UXS1	UXS1	unknown	five_prime	43600430	4	219	False	lost	0.0	False	Protein kinase domain			{TRIM24:RET} RET (NM_020975) - UXS1 (NM_001253875) rearrangement: t(2;10)(q12;q11.21)(chr2:g.106728260::cl			
EVT-P-0029222-T03-IM6-224	P-0029222-T03-IM6		RUFY1-RET	RUFY1	unknown	three_prime	43610037	11	663	False	retained	1.0	False	Protein kinase domain			mid-exon {RET:UXS1} RUFY1 (NM_025158) - RET (NM_020975) fusion: t(5;10)(q35;q11.21)(chr5:g.179025890::cl			

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint	breakpoint	breakpoint	protein	protein	breakpoint	main_ref	chr	start	end	domain	disrupted	description	annotation	effect_on
EVT-P-0031340-T02-IM6-225	P-0031340-T02-IM6		RUFY2-RET	RUFY2	in-frame	three_prime	43611559	12	713	True	retained	1.0	False	Protein kinase domain					RUFY2 (NM_017987) - RET (NM_020975) fusion: c.1045-640:RUFY2_c.2137-473:RETinv Protein Fusion: in frame {RUFY2:RET} SPECC1L NA (NM_001145468) - RET (NM_020975) fusion (SPECC1L exons 1-10 fused to RET exons 12-20): t(10;22)(q11.21;22q11.23)(chr10:g.43610353)		
EVT-P-0026614-T01-IM6-227	P-0026614-T01-IM6		SPECC1L-RET	SPECC1L	in-frame	three_prime	43611575	12	713	True	retained	1.0	False	Protein kinase domain					fusion (SPECC1L exons 1-10 fused to RET exons 12-20): t(10;22)(q11.21;22q11.23)(chr10:g.43610353)		
EVT-P-0036400-T06-IM6-228	P-0036400-T06-IM6		SPECC1L-RET	SPECC1L	in-frame	three_prime	43610353	11	712	True	retained	1.0	False	Protein kinase domain					fusion: t(10;22)(q11.21;q11.23)(chr10:g.43610353) Protein Fusion: in frame {SPECC1L:RET} SPECC1L NA (NM_001145468) - RET (NM_020975) fusion: t(10;22)(q11.21;q11.23)(chr10:g.43610353) Protein Fusion: in frame {SPECC1L:RET} SPECC1L NA (NM_001145468) - RET (NM_020975) fusion (SPECC1L exons 1-9 fused to RET exons 12-20): t(10;22)(q11.21;q11.23)(chr10:g.43610353) Protein Fusion: in frame {SPECC1L:RET} SPECC1L NA (NM_001145468) - RET (NM_020975) fusion (SPECC1L exons 1-9 fused to RET exons 12-20): t(10;22)(q11.21;q11.23)(chr10:g.43610353) Protein Fusion: in frame {SPECC1L:RET}		
EVT-P-0036400-T04-IM6-229	P-0036400-T04-IM6		SPECC1L-RET	SPECC1L	in-frame	three_prime	43610353	11	712	True	retained	1.0	False	Protein kinase domain					fusion (SPECC1L exons 1-9 fused to RET exons 12-20): t(10;22)(q11.21;q11.23)(chr10:g.43610353) Protein Fusion: in frame {SPECC1L:RET} SPECC1L NA (NM_001145468) - RET (NM_020975) fusion (SPECC1L exons 1-9 fused to RET exons 12-20): t(10;22)(q11.21;q11.23)(chr10:g.43610353) Protein Fusion: in frame {SPECC1L:RET}		
EVT-P-0036400-T03-IM6-230	P-0036400-T03-IM6		SPECC1L-RET	SPECC1L	in-frame	three_prime	43610353	11	712	True	retained	1.0	False	Protein kinase domain					fusion (SPECC1L exons 1-9 fused to RET exons 12-20): t(10;22)(q11.21;q11.23)(chr10:g.43610353) Protein Fusion: in frame {SPECC1L:RET} SPECC1L NA (NM_001145468) - RET (NM_020975) fusion (SPECC1L exons 1-9 fused to RET exons 12-20): t(10;22)(q11.21;q11.23)(chr10:g.43610353) Protein Fusion: in frame {SPECC1L:RET}		

3 Figures

Figure 1: domain retention outliers

partner block → breakpoint → retained RET portion (domain-colored, exon ticks)



Figure 2: fusion schematic

Reference vs msk_impact_50k_2026



Figure 3: reference comparison